Beyond the Symmetric Thin Ship: Asymmetry, Centreplane Lifting, and Heel

(Part II of the `michell` calculation notes)

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Abstract

Part I documented the classical Michell wave-resistance integral for a hull that is symmetric about its centerplane: a thin *thickness* distribution radiating a *source* wave system, evaluated exactly per B-spline knot span. This second note documents the three extensions built on top of that kernel, all of which reuse the Part I machinery essentially unchanged. First, a *port/starboard-asymmetric* hull is decomposed into a symmetric thickness part and an antisymmetric *camber* part; the camber radiates a centreplane *y-dipole* wave system that superposes on the source system without a cross term. Second, the dipole *magnitude* — which no pointwise boundary condition fixes, because the doublet density is a non-local functional of the camber slope — is obtained honestly by *solving* a centreplane lifting-surface problem with a horseshoe vortex lattice, replacing the crude strip closure $\mu = 2Uf_a$. Third, a *heeled* hull is handled by tilting the source centreplane, which turns the vertical decay rate *complex* and otherwise leaves the exact per-span kernel untouched. Throughout, section references point at the Rust source. All notation and constants are inherited from Part I; in particular $\nu = g/U^2$, $k_x = \nu\lambda$, $\kappa = \nu\lambda^2$, and $F(\lambda) = I + \mathrm{i}J$ is the source free-wave amplitude of Part I, Eq. (inner).

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1 Why the symmetric integral is not enough

The classical Michell integral of Part I represents the hull by a single-valued half-breadth $f(x, z) \geq 0$ and a sheet of *sources* of strength $\sigma = 2U\, \partial f/\partial x$ on the centerplane. That model is blind to three physically important situations:

1. **Port/starboard asymmetry.** A hull whose two sides differ — a cambered demihull, an asymmetric catamaran float, a hull at a small drift (yaw) angle — carries a transverse *lift* in addition to its thickness. Its wave system has an antisymmetric component that a single half-breadth cannot express.

2. **The lifting closure.** The strength of the antisymmetric (dipole) system is *not* fixed pointwise by the body boundary condition, unlike the source strength. Getting its magnitude right requires solving a lifting-surface integral equation.

3. **Heel.** A hull rolled about its longitudinal axis is asymmetric *relative to the horizontal free surface*, even if it is geometrically symmetric about its own centreplane. Its keel swings off the earth-vertical plane and it can no longer be written as port/starboard half-beams there.

The unifying idea is that all three are small perturbations that enter through the *same* inner integral and the *same* outer θ-quadrature as Part I. The only new ingredients are (a) a second coefficient field for the antisymmetric slope, (b) a real spectral *weight* coupling that field to a transverse-wavenumber factor, (c) a linear solve for the lifting density, and (d) a complex generalisation of the vertical-decay moment. Table 1 maps each extension to its entry point in the crate.

Extension	Entry point	Kernel change
Asymmetry (strip closure)	`multihull_wave_resistance_with`	second slope field $\partial f_a/\partial x$
Asymmetry (solved lift)	`asymmetric_wave_resistance_lifting`	centreplane vortex lattice
	`multihull_wave_resistance_lifting`	
Heel	`heel_wave_resistance`	complex vertical decay κ

Table 1: The three extensions and where they attach to the Part I kernel (`michell.rs`).

2 Decomposition of an asymmetric hull

An asymmetric hull is built from *two* half-breadth surfaces (`Hull::new_asymmetric`, `hull.rs`): a starboard side $y = +f_+(x, z) \geq 0$ and a port side $y = -f_-(x, z) \leq 0$, both stored as non-negative magnitudes. The two surfaces must share identical degrees and knot vectors — only their control nets may differ — so that both live on the same grid of knot spans. On that shared parametrisation the hull splits linearly into a *symmetric* (thickness) part and an *antisymmetric* (camber) part,

$$f_{\mathrm{sym}}(x, z) = \tfrac{1}{2}\big(f_+ + f_-\big), \qquad f_a(x, z) = \tfrac{1}{2}\big(f_+ - f_-\big). \tag{1}$$

Because a B-spline is linear in its control net, (1) is applied directly to the control points: $P^{\mathrm{sym}}_{ij} = \frac{1}{2}(P^+_{ij} + P^-_{ij})$ and $P^a_{ij} = \frac{1}{2}(P^+_{ij} - P^-_{ij})$. The symmetric net is itself a valid non-negative half-breadth (a mean of two non-negative nets), so the base hull is constructed from it and *every* validated symmetric quantity of Part I — volume, LCB/VCB, the source slope field c_{ab} — is reused verbatim. The antisymmetric net produces a second per-span slope field

$$\frac{\partial f_a}{\partial x}(x,z) = \sum_{a,b} c^a_{ab}\,(x-x_0)^a(z-z_0)^b, \qquad (\texttt{fx\_a\_coeff}), \tag{2}$$

computed by exactly the same `compute_fx_coeff` routine as the source field, and stored with the identical flattened layout. When $f_+ \equiv f_-$ the antisymmetric net is zero and the hull is bit-for-bit the symmetric `Hull::new` of Part I.

Two integrated quantities do *not* reduce to twice the mean and are corrected for the two sides separately: the wetted surface area and the centreplane transverse (waterplane) inertia both see f_+ and f_- individually. Volume, LCB/VCB, and waterplane area depend only on the sum $f_+ + f_- = 2f_{\text{sym}}$ and are already correct from the base hull.

3 The centreplane dipole wave system

3.1 Source plus dipole

The symmetric part f_{sym} radiates the classical source system with the Part I free-wave amplitude

$$F(\lambda) = \iint \frac{\partial f_{\text{sym}}}{\partial x}\, e^{-\kappa z}\, e^{\,\mathrm{i}k_x x}\,\mathrm{d}x\,\mathrm{d}z, \qquad k_x = \nu\lambda,\ \kappa = \nu\lambda^2. \tag{3}$$

The antisymmetric part f_a is a *camber* distribution on the centreplane. Physically it is a sheet of transverse (y-normal) *doublets* of density $\mu(x,z)$: the potential jump across the centreplane. Such a doublet sheet radiates a free wave carrying *one extra factor of the transverse wavenumber* $k_y = \nu\lambda\sqrt{\lambda^2-1}$ relative to a source of the same integrated strength,

$$A_d(\theta) = \mathrm{i}\,k_y \iint \mu(x,z)\, e^{-\kappa z}\, e^{\,\mathrm{i}k_x x}\,\mathrm{d}x\,\mathrm{d}z. \tag{4}$$

Everything downstream is a matter of fixing μ and reducing (4) to a weight on a familiar integral.

3.2 The strip closure and the real dipole weight

The crudest way to fix the density is the *strip closure* $\mu = 2Uf_a$ — the direct antisymmetric analogue of Michell's source strength $\sigma = 2U\,\partial f_{\text{sym}}/\partial x$. Substituting it into (4) and integrating by parts in x, using that f_a closes (vanishes) at bow and stern so the boundary term drops,

$$\iint f_a\, e^{-\kappa z} e^{\,\mathrm{i}k_x x}\,\mathrm{d}x\,\mathrm{d}z = \frac{\mathrm{i}}{\nu\lambda} \underbrace{\iint \frac{\partial f_a}{\partial x}\, e^{-\kappa z} e^{\,\mathrm{i}k_x x}\,\mathrm{d}x\,\mathrm{d}z}_{G(\lambda)}, \tag{5}$$

which introduces the companion amplitude $G(\lambda)$ — the *same* inner integral as the source, but over the antisymmetric slope field (2). The two factors of i (one from the dipole, one from the integration by parts) collapse to a *real* weight in this crate's amplitude units (dividing by $2U$),

$$\boxed{A_d(\lambda) = -\,w(\lambda)\,G(\lambda), \qquad w(\lambda) = C_{\text{dip}}\sqrt{\lambda^2-1}.} \tag{6}$$

This is `dipole_weight` in the source; the closure constant $C_{\text{dip}} = 1$ (`DIPOLE_WEIGHT_C`) is the value self-consistent with the source normalisation under the strip closure. The weight vanishes as $\lambda \to 1$ ($\theta \to 0$): a wave running dead ahead carries no transverse wavenumber and is blind to side-to-side asymmetry, so the whole dipole contribution is fed by the diverging (large-λ) part of the spectrum.

Warning on magnitude. The *structure* of (6) — the $\sqrt{\lambda^2-1}$ weight, the odd-in-θ parity, the additive separation below — is exact. The overall *constant* C_{dip} is not. Unlike the source strength, which the boundary condition fixes pointwise (a source sheet's normal-velocity *jump* equals its local strength), the doublet density is fixed by the *mean* of the two-sided normal velocities, and a doublet sheet's mean normal velocity is a hypersingular (finite-part) integral of μ. So μ is genuinely *non-local* in $\partial f_a/\partial x$: no exact pointwise closure exists, and $\mu = 2Uf_a$ is only the crudest strip estimate. Getting the magnitude right is the subject of Section 4.

3.3 Additive resistance and the θ-parity argument

Combining the two systems, hull member j (at fleet offset Δx_j, transverse position y_j) contributes to the $\pm\theta$ wave systems the amplitudes

$$A_{\pm,j} = \big(F_j \mp w\,G_j\big)\,\exp\!\big[\mathrm{i}\,(k_x\Delta x_j \pm k_y y_j)\big], \tag{7}$$

and the resistance integrand is the mean of the two signed systems, exactly as for a symmetric fleet in Part I:

$$R_{\text{w}} = \frac{4\rho g^2}{\pi U^2}\int_0^{\pi/2} \tfrac{1}{2}\big(|A_+|^2 + |A_-|^2\big)\,\sec^3\theta\,\mathrm{d}\theta, \tag{8}$$

with $A_\pm = \sum_j A_{\pm,j}$ (`amp_sq` in `multihull_wave_resistance_with`). The source amplitude F is *even* in θ and the dipole term wG is *odd* (the $+\theta$ system carries $-wG$, the $-\theta$ system $+wG$). Their cross term therefore integrates to zero over the symmetric Kelvin fan, and for a single hull (8) separates cleanly:

$$\boxed{R_{\text{w}} = R_{\text{source}}(f_{\text{sym}}) + R_{\text{dipole}}(f_a).} \tag{9}$$

A symmetric hull ($f_a \equiv 0$) recovers classical Michell exactly. For a fleet the dipole amplitudes superpose and interfere with the source amplitudes through the placement phases, so an asymmetric-demihull catamaran's asymmetry and its demihull interference are handled together in one spectrum.

Kernel cost. The source and dipole amplitudes share a single pass of the z-moments (Part I, Eqs. (N)): only the x-span accumulation is repeated against the second coefficient array. `InnerIntegral::eval_pair` returns (F, G) together, so an asymmetric evaluation costs barely more than a symmetric one.

4 The centreplane lifting solve ("level 2")

To replace the strip constant with a physically grounded density we *solve* the lifting-surface problem for $\mu(x,z)$. The centreplane $x \in [0,L]$, $z \in [0,T]$ carries the camber surface f_a, whose streamwise slope $\partial f_a/\partial x$ is the downwash the transverse doublet sheet must produce — the direct analogue of a wing's camber-slope condition.

4.1 Two-dimensional proving ground

Before the delicate 3D free-surface solve, the crate builds and validates its 2D ancestor — classical thin-airfoil theory (`lifting.rs`) — which exercises the same machinery: discretising a lifting sheet, imposing the Kutta condition, and recovering the loading. A thin airfoil of unit chord with camber-line slope $\mathrm{d}\eta/\mathrm{d}x$ is represented by a bound-vortex sheet $\gamma(x)$ satisfying the Cauchy singular integral equation

$$\frac{1}{2\pi}\ \mathrm{p.v.}\!\int_0^1 \frac{\gamma(\xi)}{x-\xi}\,\mathrm{d}\xi = U\Big(\alpha - \frac{\mathrm{d}\eta}{\mathrm{d}x}(x)\Big). \tag{10}$$

Two independent solvers cross-check each other and closed-form results:

- **Glauert Fourier series** (`glauert`), with $x = \frac{1}{2}(1-\cos\theta)$ and $\gamma/U = 2\big[A_0\frac{1+\cos\theta}{\sin\theta} + \sum_{k\geq 1} A_k \sin k\theta\big]$; the $\sin k\theta$ terms vanish at both edges so the Kutta condition is built in, and $C_L = \pi(2A_0 + A_1)$. A flat plate gives $A_0 = \alpha$, hence the canonical $C_L = 2\pi\alpha$.

- **Lumped-vortex lattice** (`solve_vortex_lattice`) with the Pistolesi $\frac{1}{4}$–$\frac{3}{4}$ rule: a point vortex at each panel's quarter-point, flow tangency collocated at its three-quarter point, which enforces the Kutta condition automatically. This is the desingularised 2D form of the hypersingular kernel the 3D solve must integrate.

In hull terms the camber line is the antisymmetric half-beam f_a at a waterline strip and the forcing is $-U\,\partial f_a/\partial x$ (no separate incidence). The solved loading integrates to the doublet density $\mu(x) = \int_0^x \gamma\,\mathrm{d}\xi$ — exactly the quantity the free-wave amplitude carries.

4.2 Three-dimensional horseshoe vortex lattice

The 3D engine (`lifting3d.rs`) generalises the lattice: each panel carries a *horseshoe vortex* — a bound segment at the quarter-chord plus two trailing legs running downstream to $+\infty$ along $\hat{x}$ — and flow tangency is collocated at the three-quarter-chord point. The trailing legs make the Kutta condition automatic. Assembling the normal-wash influence coefficients $A_{ij} = (\mathbf{v}_{j\to i}\cdot\hat{n}_i)$ from the Biot–Savart law (finite-segment and semi-infinite closed forms, after Katz & Plotkin) and solving

$$A\,\Gamma = b, \qquad b_i = -\,\mathbf{U}_\infty\cdot\hat{n}_i, \tag{11}$$

gives the bound circulation Γ on every panel. Lift follows from Kutta–Joukowski, $C_L = 2\sum_j \Gamma_j\,\Delta y_j/S$. The engine is validated on a flat rectangular wing against the two exact asymptotic limits that bracket the lift-curve slope: $\mathrm{d}C_L/\mathrm{d}\alpha \to 2\pi$ as $\mathrm{AR}\to\infty$ (2D limit) and $\to \frac{\pi}{2}\mathrm{AR}$ as $\mathrm{AR}\to 0$ (R.T. Jones slender-wing limit), with the Prandtl lifting-line estimate $2\pi/(1+2/\mathrm{AR})$ in between.

4.3 Free surface as a rigid wall (double body)

The hull-specific step (`centerplane.rs`) applies this lattice to the *vertical* centreplane. In the low-Froude near field the free surface $z=0$ behaves as a rigid wall ($\partial\phi/\partial z = 0$), consistent with how Michell's source strength is set on the double-body flow. This is enforced by **reflecting the centreplane about** $z = 0$ with the same circulation sign, so the bound vortices run continuously through the waterline. The consequences:

- the waterline acts as a wing *root* (maximum loading) and the keel $z = T$ as a free *tip* (loading relieved to zero);

- a surface-piercing lifting surface of draft T therefore behaves as a wing of span $2T$ — the classical *effective-aspect-ratio doubling*, checked in the tests against a full wing of span $2T$ to $\sim 0.1\%$ at $\alpha = 3^\circ$.

The camber enters through the panel *normal* $\hat{n} = (-f_{a,x}, 1, 0)/\sqrt{1+f_{a,x}^2}$, so the freestream stays $(1,0,0)$ and the solve is called at zero incidence. Solving for Γ and accumulating from the leading edge gives the doublet density $\mu(x,z) = \int \gamma$ on the physical half $z \geq 0$, together with a side-force coefficient $C_Y = 2\sum_{\text{phys}} \Gamma\, \Delta z/(LT)$.

4.4 From the solved density to the wave amplitude

The solved density feeds the dipole free-wave amplitude by midpoint quadrature over the panel grid (`doublet_free_wave_amplitude`, each sample carrying cell area $\Delta x\, \Delta z$),

$$G_\mu(\lambda) = \iint \frac{\mu(x,z)}{U}\, e^{-\kappa z}\, e^{\,\mathrm{i}k_x x}\, \mathrm{d}x\, \mathrm{d}z, \tag{12}$$

from which the dipole amplitude is

$$\boxed{A_d(\lambda) = \mathrm{i}\, \tfrac{1}{2}\, k_y\, G_\mu(\lambda), \qquad k_y = \nu\lambda\sqrt{\lambda^2-1}.} \tag{13}$$

Note that here G_μ integrates μ *directly* (not the slope), so no integration by parts is applied and the amplitude stays imaginary — contrast the strip form (6), where the by-parts step made it real. The two paths share *one* normalisation: substituting the strip closure $\mu/U = 2f_a$ into (13) and integrating by parts via (5) reproduces $A_d = -\sqrt{\lambda^2-1}\, G$ exactly. The factor $\frac{1}{2}$ is precisely what makes the closure $\mu/U = 2f_a$ collapse onto the $C_{\text{dip}} = 1$ weight. The solved μ has a different chordwise/vertical *shape* than $2f_a$, giving a dipole of the same order but a speed-dependent ratio; for a 2D flat plate $\mu_{\text{lift}}/\mu_{\text{strip}} = \pi/2$.

The single-hull entry point `asymmetric_wave_resistance_lifting` and the fleet form `multihull_wave_resistance_lifting` solve each asymmetric member's centreplane *once* (up front, not per λ), then superpose the source amplitude F_j and the solved dipole $A_{d,j}$ with the member's placement phase, exactly as in (8). The dipole phase is re-centred to the hull midpoint (the solver works on $x \in [0, L]$) so it shares the source's origin. This is the far-field coupling: each member's lifting problem is solved independently with its own free-surface image, and their dipole *wave* systems interfere through the superposition — so an asymmetric-demihull catamaran's demihull interference is captured. Near-field lifting cross-induction between close demihulls is *not* (that would need one coupled lifting solve over all centreplanes at once). The default grid is $n_x = 48$ streamwise $\times$ $n_z = 16$ vertical panels (`LiftingGrid`).

5 Heeled hulls

A hull heeled by an angle φ about its longitudinal axis is asymmetric relative to the horizontal free surface, but its keel swings off the earth-vertical centreplane, so it cannot be written as port/starboard half-beams there. Thin-ship theory instead keeps the sources on the ship's *own* (now tilted) centreplane: a strip at ship-depth z sits at earth depth $z\cos\varphi$ and transverse offset $-z\sin\varphi$. The transverse offset contributes a transverse-wavenumber phase $e^{-\mathrm{i}k_y\,(-z\sin\varphi)}$ to the kernel, which combines with the earth-vertical decay $e^{-\nu\lambda^2 z\cos\varphi}$ into a single **complex vertical decay**

$$\boxed{\kappa = \nu\lambda^2\cos\varphi \;+\; \mathrm{i}\,\nu\lambda\sqrt{\lambda^2-1}\,\sin\varphi \;=\; \nu\lambda^2\cos\varphi + \mathrm{i}\,k_y\sin\varphi.} \tag{14}$$

The imaginary part is the same transverse-wavenumber coupling as the asymmetric dipole, arising here purely from geometry.

This is the *only* change to the Part I kernel. The x-oscillation moments M_a and the per-span accumulation are identical; only the z-moment $N_b = \int_0^{h_z} Z^b e^{-\kappa Z}\,\mathrm{d}Z$ becomes complex, evaluated by `exp_moments_complex` — the *same* series and integration-by-parts recurrence as the real routine (Part I, Eqs. (Nrec)–(Nseries)), promoted to $\mathbb{C}$ with $\operatorname{Re}\kappa \geq 0$ so the decay never grows. The real path is left entirely untouched, so $\varphi = 0$ reproduces `wave_resistance` exactly (`HeelInner` in `michell.rs`).

The tilt makes the amplitude F depend on the *sign* of θ (through the $\pm$ in $k_y \sin\varphi$), so the resistance averages the two signed systems,

$$R_{\mathrm{w}} = \frac{4\rho g^2}{\pi U^2} \int_0^{\pi/2} \tfrac{1}{2}\left(|F_+|^2 + |F_-|^2\right) \sec^3\theta \,\mathrm{d}\theta, \tag{15}$$

where $F_\pm$ uses κ with $\pm \mathrm{i} k_y \sin\varphi$. The result is *even* in φ: port and starboard heel are mirror images. Because the tilt spreads the centreplane transversely by up to $T|\sin\varphi|$, the outer-quadrature panels are sized with an effective transverse half-extent $y_{1/2} = T|\sin\varphi|$ to resolve the heel-induced oscillation.

This captures the asymmetric *wave-making* of the tilted thickness distribution — the leading heel effect. It does *not* include the lifting side-force that a heeled-and-yawed (drifting) hull develops; that would enter as a separate forcing into the centreplane lifting solve of Section 4.

6 Data-flow summary

	Asymmetry — strip closure	
two nets $P_{ij}^{\pm}$	$\longrightarrow$	f_{sym}, f_a (Eq. 1); slope fields c_{ab}, c_{ab}^a
c_{ab}, c_{ab}^a	$\longrightarrow$	(F, G) together (`eval_pair`)
G, weight $w = \sqrt{\lambda^2 - 1}$	$\longrightarrow$	dipole $A_d = -wG$ (Eq. 6)
$\frac{1}{2}(\lvert F - wG\rvert^2 + \lvert F + wG\rvert^2)$	$\longrightarrow$	outer θ-quadrature $\Rightarrow$ $R_{\mathrm{w}} = R_{\mathrm{src}} + R_{\mathrm{dip}}$
	Asymmetry — solved lift	
$\partial f_a/\partial x$ on centreplane	$\longrightarrow$	vortex-lattice solve Γ (double body)
Γ	$\longrightarrow$	doublet density $\mu = \int \gamma$, side force C_Y
μ	$\longrightarrow$	G_μ; dipole $A_d = \mathrm{i}\frac{1}{2}k_y G_\mu$ (Eq. 13)
$F_j + A_{d,j}$, placements	$\longrightarrow$	outer θ-quadrature $\Rightarrow R_{\mathrm{w}}$
	Heel	
heel φ	$\longrightarrow$	complex κ (Eq. 14)
κ	$\longrightarrow$	complex z-moments N_b (`exp_moments_complex`)
$\frac{1}{2}(\lvert F_+\rvert^2 + \lvert F_-\rvert^2)$	$\longrightarrow$	outer θ-quadrature $\Rightarrow R_{\mathrm{w}}$ (even in φ)

All three extensions share the exact per-span inner kernel and the single smooth outer θ-quadrature of Part I. The only approximations added are the strip closure constant C_{dip} (Section 3.2, removed by the lifting solve), the panel discretisation of the lifting solve and its midpoint quadrature (Section 4), and — as in Part I — the one-dimensional angular quadrature. The wave-pattern (wake) branch of Part I is unchanged: the free-wave elevation grid is still built from the symmetric source amplitudes.

References. Tuck (1989); Dambrine, Pierre & Rousseaux (2016); Tuck, Scullen & Lazauskas (2001, 2002); Katz & Plotkin, *Low-Speed Aerodynamics* (horseshoe-lattice `VORTXL`); Glauert, *The*

Elements of Aerofoil and Airscrew Theory; Kaklis & Papanikolaou (21st Symp. Naval Hydro., Appendix A, on the hypersingular doublet closure).